

A Performance-Portable Particle–Mesh Framework for Cosmological Structure Formation with IPPL

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Abstract

We present the first cosmological structure-formation application built on the Independent Parallel Particle Layer (IPPL), a C++ framework for distributed, performance-portable particle–mesh algorithms. The present model evolves one collisionless cold-matter species in a periodic flat Λ CDM universe. It combines Gaussian first-order Lagrangian perturbation theory initial conditions, cloud-in-cell particle–mesh coupling, a distributed spectral Poisson solve, and kick–drift–kick integration in the scale factor. The implementation inherits the MPI, Kokkos, and heFFTe abstractions previously exercised by IPPL’s plasma particle-in-cell mini-applications, while adding cosmological background evolution, transfer functions, reproducible Fourier-space initialization, and cosmology-specific validation.

The validation establishes linear evolution on one to four MPI ranks with an OpenMP backend and matched initial-condition statistics against an independent generator. It also tests frozen forces, analytical pre-crossing trajectories, and nonlinear evolution against native plain-PM FastPM. Matched-discretization agreement is close, but crossed particle/mesh studies retain resolution failures; the completed Gaussian matrix does not yet establish a qualified resolution band. These bounded results position the code as a portable particle–mesh research and verification platform, not a replacement for mature production cosmology codes. Higher-resolution studies, second-order initial conditions, short-range gravity, scalable output, and accelerator and multi-node qualification remain necessary for broader scientific and performance claims.

Scope and contribution. The contribution is twofold. First, it translates a validated particle–mesh software stack from plasma physics to cosmological gravity without duplicating the portability layer. Second, it defines a staged validation path in which background cosmology, initial conditions, discretized force, time integration, parallel invariance, and performance are tested independently before nonlinear science claims are made. The initial linear results were summarized at development snapshot `0f2953bde`; Appendix A records the completed validation campaigns through 4 October 2026, their separate provenance, and their retained failures. Results marked [TODO: measurement] are planned rather than completed.

Position in computational cosmology

Cosmological simulation is a mature field with several distinct algorithmic families. GADGET-4 combines tree or fast-multipole short-range gravity with a particle–mesh long-range solver and includes hydrodynamics, on-the-fly structure finding, and second-order initial conditions [1]. SWIFT uses fine-grained tasking and couples fast-multipole gravity to an optional Fourier particle–mesh solver, alongside hydrodynamics and galaxy-formation models [2]. HACC was designed for extreme-scale cosmological simulations and uses an architecture-dependent short-range solver around a

common long-range spectral method [3]. Abacus attains high force accuracy and throughput with a specialized near/far decomposition; the AbacusSummit campaign evolved roughly 60 trillion particles over many cosmologies [4]. Mesh and adaptive-mesh codes such as RAMSES and Nyx occupy a complementary Eulerian niche, particularly when gas dynamics and refinement are central [5, 6].

Pure and approximate particle–mesh methods are an equally important comparison. FastPM modifies the kick and drift operators to preserve linear displacement growth with few time steps and targets fast mock-catalog generation, while COLA evolves residual motion around perturbative trajectories for a similar purpose [7, 8]. CONCEPT couples a PM long-range solver to a short-range particle–particle method and adaptive stepping, with tight integration of cosmological initial conditions and analysis products [9]. Thus neither particle–mesh cosmology nor high-throughput ensemble generation is by itself a novelty claim for this work. The research question is instead whether a reusable particle-and-field framework can retain cosmological accuracy and useful efficiency while holding the high-level implementation fixed across vendor architectures.

Against these production systems, the near-term role of an IPPL-based code is not feature parity. Its distinctive value is a compact, open particle–mesh test bed in which the same application-level C++ code targets multiple CPU and GPU programming backends through Kokkos, distributed FFTs through heFFTe, and domain decomposition through IPPL. This makes it suitable for controlled studies of portable deposition and interpolation kernels, FFT-based gravity, reproducible distributed initialization, load balancing, mixed precision, and new Poisson or time-integration algorithms. The approach mirrors the role of the ALPINE plasma mini-apps as an interface between physics, algorithms, and computer architecture [10].

Table 1 summarizes the intended positioning. The immediate scientific target is periodic collisionless large-scale structure in the linear and mildly nonlinear regimes. High-dynamic-range halo and galaxy formation requires a short-range force method and, for baryonic science, hydrodynamics and subgrid physics. Those additions should be motivated by a specific science case rather than treated as prerequisites for a useful particle–mesh research code.

1 Cosmology model

1.1 Background expansion

The current model is a spatially flat, radiation-free Λ CDM cosmology with one collisionless matter species. The scale factor a is normalized to unity today and is related to redshift by $a = (1 + z)^{-1}$. With $\Omega_\Lambda = 1 - \Omega_m$, the dimensionless Hubble function is

$$E(a) \equiv \frac{H(a)}{H_0} = \left(\frac{\Omega_m}{a^3} + \Omega_\Lambda \right)^{1/2}. \quad (1)$$

Radiation, curvature, massive neutrinos, evolving dark energy, and modified gravity are not included in the present validation envelope.

Positions $\mathbf{x}$ are comoving and expressed in h^{-1} Mpc. We introduce the dimensionless time $\tau = H_0 t$ and canonical momentum

$$\mathbf{p} = a^2 \frac{d\mathbf{x}}{d\tau}. \quad (2)$$

The physical peculiar velocity is then $\mathbf{v}_{\text{pec}} = 100 \mathbf{p}/a$ in km s^{-1} for coordinates in h^{-1} Mpc.

Table 1: A functional comparison that clarifies the intended role of the IPPL cosmology application. “Planned” denotes work not qualified by the results in this paper.

Capability		Mature production landscape	IPPL cosmology status and role
Periodic gravity	long-range	Spectral PM is common in HACC, TreePM/FMM-PM codes, and other large-volume solvers	Implemented with CIC and a distributed spectral Poisson solve; primary current capability
Short-range force		Trees, FMM, P ³ M, AMR, or specialized near-field algorithms	Not implemented; required for halo-scale accuracy
Initial conditions		2LPT and external production generators are standard	Deterministic distributed 1LPT, BBKS and tabulated transfer functions; 2LPT is planned
Heterogeneous hardware	hard-	Often optimized through code-specific CPU/GPU paths	Single-source Kokkos abstraction inherited from IPPL; accelerator cosmology runs remain to be qualified
Baryonic physics		Available in codes such as SWIFT, GADGET-4, Nyx, and RAMSES	Outside the present paper; possible through future IPPL fields or coupling to a hydrodynamics package
Research use		Precision survey production and astrophysical prediction	Portable algorithms, verification, performance studies, and rapid particle-mesh method development

1.2 Collisionless particle-mesh equations

Let $\delta = \rho/\bar{\rho} - 1$ be the matter density contrast. We define a rescaled potential ϕ_0 through

$$\nabla^2 \phi_0 = \frac{3}{2} \Omega_m \delta, \quad \mathbf{F} = -\nabla \phi_0. \quad (3)$$

Using a as the independent variable gives

$$\frac{d\mathbf{p}}{da} = \frac{\mathbf{F}}{a^2 E(a)}, \quad \frac{d\mathbf{x}}{da} = \frac{\mathbf{p}}{a^3 E(a)}. \quad (4)$$

The periodic zero mode is explicitly removed. Equations (3) and (4) are discretized with a uniform cell-centered mesh, equal-mass macro-particles, and cloud-in-cell (CIC) assignment. The same shape function is used for scatter and gather, which avoids an asymmetric particle-mesh coupling.

One simulation step is a kick-drift-kick composition. For a force held fixed during a kick and a momentum held fixed during a drift, the cosmological factors are

$$K(a_0, a_1) = \int_{a_0}^{a_1} \frac{da}{a^2 E(a)}, \quad D_{\text{drift}}(a_0, a_1) = \int_{a_0}^{a_1} \frac{da}{a^3 E(a)}. \quad (5)$$

Steps are uniform in $\ln a$, with the geometric midpoint separating the two kicks. Host-side adaptive quadrature evaluates Eq. (5); particle and mesh kernels remain in the selected Kokkos execution space. The scheme is second order in time. Mechanical energy is not expected to remain constant in an expanding background; validation instead uses the linear growing mode, convergence, mass conservation, and parallel consistency.

1.3 Current physical limits

The mesh supplies the only force resolution. There is no CIC-window deconvolution, sub-cell short-range force, adaptive mesh, or independent force softening parameter. The present application is therefore well matched to linear modes and large-volume algorithm studies but not yet to precision halo internal structure. A credible nonlinear extension should add and separately validate one of the following: P³M, TreePM/FMM-PM, or a refined mesh. This choice affects conservation, scaling, load balance, and portability and should be treated as a central research decision.

2 Initial conditions

2.1 Linear spectrum and growth

The $z = 0$ linear matter spectrum is

$$P(k) = Ak^{n_s}T^2(k), \quad (6)$$

with k in $h \text{ Mpc}^{-1}$ and P in $(h^{-1} \text{ Mpc})^3$. The amplitude A is fixed by

$$\sigma_8^2 = \frac{1}{2\pi^2} \int dk k^2 P(k) W^2(8k), \quad W(x) = 3 \frac{\sin x - x \cos x}{x^3}. \quad (7)$$

The implementation supports the BBKS analytic CDM transfer function and a CMBFAST-style table containing cold-dark-matter and baryon transfer columns. For the table option, the total matter transfer function is formed with the configured baryon fraction and normalized at low k . The input table must cover the largest three-dimensional mesh wave number; extrapolation beyond the last tabulated wave number is rejected.

The growing solution is evaluated as

$$D_{\text{raw}}(a) = \frac{5\Omega_m}{2} E(a) \int_0^a \frac{da'}{a'^3 E^3(a')}, \quad D(a) = \frac{D_{\text{raw}}(a)}{D_{\text{raw}}(1)}, \quad (8)$$

with logarithmic growth rate $f = d \ln D / d \ln a$. This integral solution is cross-checked against an independent growth-ODE calculation.

2.2 Distributed Gaussian 1LPT

Particles begin on a cell-centered Lagrangian lattice $\mathbf{q} = (\mathbf{i} + 1/2)L/N$. Independent Gaussian Fourier coefficients obey

$$\langle |\hat{\delta}(\mathbf{k}, a)|^2 \rangle = \frac{D^2(a)P(k)}{L^3}, \quad (9)$$

under the transform convention used by the implementation. Hermitian partners are constructed explicitly; the DC mode and initial-condition Nyquist planes are zero. First-order Lagrangian perturbation theory gives

$$\hat{\psi}_0(\mathbf{k}) = \frac{i\mathbf{k}}{k^2} \hat{\delta}(\mathbf{k}, a = 1), \quad (10)$$

$$\mathbf{x}(a_i) = \mathbf{q} + D(a_i)\psi_0, \quad (11)$$

$$\mathbf{p}(a_i) = a_i^2 E(a_i) f(a_i) D(a_i) \psi_0. \quad (12)$$

Random values are generated from a configured 64-bit seed and a canonical global Fourier-mode index. Consequently, a mode receives the same coefficient for different MPI decompositions and

OpenMP schedules. This is stronger than merely using the same seed on every rank and enables direct phase-space comparisons across rank counts.

The application also provides a single sinusoidal mode and a uniform lattice. The former isolates force and growth errors; the latter tests the periodic zero-force equilibrium. Production cosmology will require second-order Lagrangian perturbation theory to reduce transients and a documented path for ingesting community initial-condition formats.

3 A performance-portable implementation

3.1 Reuse of the IPPL particle–mesh stack

The implementation follows the software decomposition established for IPPL’s plasma particle-in-cell applications [10]. MPI distributes fields and particles across ranks. Kokkos views hold field and particle attributes and Kokkos execution policies map kernels to the selected on-node backend [11]. heFFTe supplies distributed transforms with CPU and GPU backends [12]. Particle migration uses a pack–send–receive–unpack path, while field boundary communication uses halo cells. Communication buffers can be overallocated to reduce repeated device allocation costs. Orthogonal recursive bisection is available when particle clustering produces load imbalance.

This decomposition carries over directly to gravity:

1. CIC scatter deposits particle mass onto the distributed mesh.
2. The mean density is subtracted and the periodic Poisson equation is solved in Fourier space.
3. The differentiated potential gives the mesh force.
4. CIC gather interpolates the force to particle positions.
5. Kokkos kernels apply kick and drift operations, periodic wrapping, and particle migration.

No plasma-specific charge or Boris-push logic is reused. The reusable layer is the particle–mesh infrastructure and its parallel implementation.

3.2 Memory, execution, and reproducibility

Particle positions, canonical momenta, forces, global identifiers, and Lagrangian coordinates remain in the configured Kokkos memory space during evolution. The background integrals are scalar host calculations. A radial $P(k)$ lookup is prepared on the host and copied to execution memory for initial condition generation. In the current prototype this lookup uses $\mathcal{O}(N^2)$ replicated storage per rank, which is acceptable for local validation but should be replaced or distributed before extreme-scale runs.

MPI reductions, deposition atomics, and FFT algorithms may change floating-point summation order across decompositions and devices. Reproducibility is therefore defined in two layers: Fourier random phases and global particle identifiers are deterministic; evolved floating-point fields are required to agree within declared tolerances rather than bit for bit. Any mixed-precision study must report its effect on the power spectrum, force error, and growth in addition to speed and memory savings.

3.3 Performance methodology

The plasma study showed that useful portability evidence separates component costs rather than reporting only total runtime [10]. The cosmology campaign will therefore time CIC scatter, halo exchange, forward and inverse FFTs, spectral Green-function and gradient kernels, CIC gather, particle push, migration, load balancing, diagnostics, and output. We will report particle updates per second, mesh cells solved per second, time per step, memory per particle and cell, and the fraction of runtime in communication.

Strong-scaling tests will fix particle and mesh counts; weak-scaling tests will fix both per rank. At least one CPU and one GPU platform should be tested, and preferably two accelerator vendors, before claiming performance portability. Performance-portability efficiency should be reported relative to an optimized backend-specific baseline where available, not inferred from successful execution alone.

Table 2: Planned performance matrix. Empty cells are intentional: the present draft does not contain accelerator or multi-node cosmology results.

Platform	Backend	Scale	Required evidence
Local multicore CPU	OpenMP	1–4 MPI ranks	completed correctness baseline
[TODO: CPU cluster]	OpenMP	multi-node	strong/weak scaling and communication split
[TODO: NVIDIA system]	CUDA	multi-node	kernel, FFT, migration, and end-to-end rates
[TODO: AMD system]	HIP	multi-node	same metrics and portability comparison

4 Validation

4.1 Validation hierarchy

The validation is organized so that a failure can be attributed to one layer. Host unit tests cover configuration, Eq. (1), the growth integral and an independent ODE oracle, Einstein–de Sitter kick and drift factors, transfer interpolation, and σ_8 normalization. Distributed tests then exercise a manufactured spectral force, particle migration, and periodic wrapping on one through four MPI ranks.

End-to-end tests include:

- a uniform lattice, which must retain zero force and conserve mass and particle number;
- axis-aligned and oblique modes, which measure scale-dependent spatial error and linear growth;
- Gaussian fields, for which reconstructed Fourier coefficients test the requested spectrum, Hermiticity, longitudinal displacement, and the momentum–displacement relation;
- rank and thread comparisons sorted by deterministic particle ID;
- time-step and mesh-refinement sequences assessed independently; and
- matched initial-condition ensembles against an independent Zarija Lukić implementation in the common flat, Gaussian, radiation-free CDM subset.

The two initial-condition generators use different random-number algorithms, so equal numeric seeds do not imply equal phases. Their comparison therefore separates deterministic background, transfer, normalization, and velocity-factor checks from statistical ensemble checks. Known lattice-origin, transform-order, velocity-unit, and Nyquist conventions are handled explicitly rather than fitted.

4.2 Current linear-regime results

Table 3 records the current local qualification. The full native campaign comprised 22 simulations and 275 checks. The matched reference campaign comprised 84 runs and 1047 checks, including IPPL runs on one, two, three, and four ranks and reference runs on one, two, and four ranks. All acceptance limits were declared before the corresponding campaign and were not relaxed after execution.

Table 3: Selected measurements from the local OpenMP qualification at initial development snapshot 0f2953bde. These results establish linear-regime behavior; later nonlinear comparisons and provenance limits are documented separately in Appendix A.

Test	Measured result
Manufactured spectral force, ranks 1–4	maximum relative error approximately 2×10^{-15}
Forced migration, ranks 1–4	ownership and all IDs/attributes preserved; exact endpoints handled
Reconstructed Gaussian Fourier coefficients	relative L_2 error 4.8×10^{-12}
Gaussian low- k linear growth on a 32^3 mesh	weighted complex error 0.333%
Late- Λ single-mode growth, $z = 9$ to $z = 0$	error 0.943%
Time convergence, position / momentum	observed order 1.988 / 1.995
Mesh refinement $16^3 \rightarrow 32^3 \rightarrow 64^3$	growth-error reduction $3.899\times / 3.849\times$
Matched-IC ensemble mean $P_{\text{measured}}/P_{\text{theory}}$	IPPL: 0.995736642–0.995736648; reference: 0.998164308–0.998196423
IPPL rank agreement in matched campaign	maximum position RMS $6.502 \times 10^{-16} h^{-1}\text{Mpc}$; momentum RMS 1.339×10^{-17} in canonical units

The result most relevant to the physical model is not any single small residual, but the combination of second-order temporal convergence, improving spatial error, correct Gaussian statistics, and rank-independent phases. The remaining sub-percent growth error at finite mesh spacing is expected for CIC PM without window deconvolution or a short-range correction and decreases under mesh refinement.

4.3 Completed nonlinear comparisons and resolution controls

The staged evidence now extends beyond this initial linear baseline. Appendix A documents an independent frozen-force comparison, analytical pre-crossing pancake evolution, 32 matched nonlinear native plain-PM runs, a separate targeted timestep extension, 34 local spatial-control runs, and a fresh 18-run Gaussian study on Merlin CPUs. It provides the experiment matrices, comparison

conventions, declared criteria, figures and report inventory. The native reference uses ordinary PM evolution, not modified FastPM or COLA stepping.

The latest Gaussian study passes every paired-code, timestep, starting-redshift, MPI and integrity criterion, but retains 116 particle/mesh-resolution failures among 1,355 checks. No Gaussian shell qualifies over that full matrix. The local spatial study qualifies only its first low-mode shell for each fixture, and earlier mass-sum and timestep failures remain explicitly recorded. These results support the implementation comparison while identifying the remaining accuracy limitations; they are not halo or continuum validation.

4.4 Remaining scientific and performance validation

The paper is not complete until the following evidence is added:

1. Extend the present matched-PM comparison to larger crossed resolutions, independent realizations, converged nonlinear power spectra and halo statistics against an established production code. Published multi-code comparisons suggest percent-level power-spectrum agreement is a meaningful target over a declared resolved range [13].
2. Quantify force error versus separation and mesh resolution, including anisotropy and the effect of any future short-range solver.
3. Add a Layzer–Irvine or equivalent expanding-universe energy diagnostic, while retaining mass, momentum, and particle-count checks.
4. Validate 2LPT against an independent generator and measure transient suppression relative to 1LPT.
5. Repeat correctness tests on each advertised Kokkos backend and on at least two multi-node systems.
6. Populate Table 2 with strong/weak scaling, component profiles, memory usage, and device-to-device comparisons.
7. Replace diagnostic CSV output with scalable parallel snapshots and add restart, power-spectrum, light-cone or halo-catalog workflows appropriate to the selected science target.

Discussion and roadmap

The current prototype demonstrates that IPPL’s particle–mesh abstraction is not specific to electrostatic plasma physics. Density deposition, a global Poisson solve, field interpolation, particle migration, and load imbalance are also the central computational motifs of long-range cosmological gravity. The cosmological application can therefore reuse a portability layer whose critical components have already been studied on heterogeneous pre-exascale systems, while keeping cosmological equations and validation independent.

The most defensible research program has three stages. Stage I, represented by this draft, is a linear PM mini-app plus a staged verification suite and bounded nonlinear comparisons, with unresolved resolution sensitivities kept explicit. Stage II adds larger-resolution accuracy studies, 2LPT, scalable I/O and restart, and accelerator and multi-node qualification. Stage III is science-case dependent: a short-range gravity method for halo resolution; neutrino or modified-gravity particles for beyond-baseline cosmology; or coupling to a mesh hydrodynamics solver for baryonic applications.

This staged scope retains the value of a small portable test bed while providing a clear path toward scientifically competitive simulations.

Code and data availability

IPPL is open source at <https://github.com/IPPL-framework/ippl>. The cosmology implementation and validation artifacts described here are currently maintained on development branch `codex/cosmology-linear`; the results in Table 3 correspond to commit `0f2953bde` and its recorded development history. Appendix A.11 identifies the later campaign reports; the Merlin run used frozen source `296fd04c0`, with documentation handoff at `9506f1e52`. The manuscript includes verified figure assets and their provenance manifest, not the full particle archives. [TODO: Replace the development reference with a public release DOI, archive input files and machine-readable results, and state licenses before submission.]

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A Validation

This appendix records the evidence available through 5 October 2026. It separates verification of equations and discrete operators, agreement between independent implementations, and sensitivity to numerical resolution. These are different questions: agreement between two PM codes does not by itself establish a continuum solution, halo accuracy, or performance portability. All results below concern periodic, flat, radiation-free cosmology with one collisionless species. Cosmological initial conditions are tested only at first Lagrangian order; frozen-force fixtures need not be cosmological realizations. The numerical campaigns used an OpenMP CPU backend; no cosmological GPU, multi-node, or exascale result is claimed.

The validation was cumulative, but it was not one homogeneous experiment. Early campaigns were run locally on macOS; the Gaussian campaign G, broadband campaign ZB, and subsequent GADGET-2 comparison GT are distinct executions on the Merlin Linux login host. Repeated cases, reused snapshots, and related checks must not be counted as independent statistical samples. A *check* is a recorded acceptance predicate, not an independent measurement. Table A.1 retains the failed predicates as well as the passing ones. Completion of a campaign and acceptance of all its scientific criteria are explicitly distinguished.

Table A.1: Completed validation campaigns used in this appendix. Report keys are resolved in Section A.11. “Runs” counts executable runs, not independent realizations. The scalar probe is a separate set of numerical comparisons.

Key	Experiment	Runs	Recorded outcome
L	Linear evolution, rank/thread and refinement tests	22	275/275 checks pass
ZP	Zarija background/transfer scalar probe	—	4,655 gated comparisons pass; nine additional diagnostics are not gated
ZI	Matched-cosmology initial conditions	84	1,047/1,047 checks pass
F	Frozen forces versus native FastPM and independent operators	64	790/790 checks pass
P	Analytical pre-crossing pancake	15	363/364 pass; one strict mass-sum check fails
E	Matched nonlinear native plain-PM evolution	32	1,723/1,725 pass; two final momentum timestep checks fail
ER	Targeted 512-step extension of E	2 new	99/99 pass; original E failures are unchanged
S	Crossed particle/mesh, translation and timestep controls, local CPU	34	2,083/2,213 pass; 130 retained failures
G	Common-phase Gaussian controls, Merlin CPU	18	1,239/1,355 pass; 116 resolution failures
ZB	Broadband 1LPT evolution versus native plain-PM FastPM, Merlin CPU	4	All launches and integrity checks complete; final code differences measured, not gated

Key	Experiment	Runs	Recorded outcome
GT	Matched $z_i = 99$ evolution with GADGET-2 TreePM, Merlin CPU	1 integration	Complete from check-point/restart; 48 bins compared, no new acceptance gate

A.1 Conventions, observables, and independent checks

Lengths are comoving h^{-1} Mpc and time is $\tau = H_0 t$. Canonical momentum is $\mathbf{p} = a^2 d\mathbf{x}/d\tau$; it is neither physical peculiar velocity nor the raw velocity exported by the Zarija generator. Particle and mesh resolutions are denoted by N_P and N_M , respectively: there are N_P^3 particles and N_M^3 mesh cells, with cell width $\Delta x = L/N_M$. Particle comparisons are sorted by global ID, require complete unique IDs and finite attributes, and use the minimum-image periodic position difference. In the matched evolution studies,

$$e_x = \frac{[N_P^{-3} \sum_j |\Delta \mathbf{x}_j|^2]^{1/2}}{\Delta x}, \quad e_p = \frac{[\sum_j |\Delta \mathbf{p}_j|^2]^{1/2}}{[\sum_j |\mathbf{p}_{j,\text{ref}}|^2]^{1/2}}. \quad (\text{A.1})$$

The analytical pancake instead normalizes trajectory errors by the corresponding analytical displacement or momentum RMS; its percentages should not be confused with position errors in mesh cells.

For the direct-mode nonlinear comparisons E, S, and G, nonzero density modes are measured directly from the particle positions, independently of either force mesh:

$$\hat{\delta}_{\mathbf{n}} = \frac{1}{N_P^3} \sum_j \exp \left[-\frac{2\pi i}{L} \mathbf{n} \cdot \mathbf{x}_j \right]. \quad (\text{A.2})$$

For the vectors $\mathbf{d}_A, \mathbf{d}_B$ of coefficients in a common mode set, define $S_A = \|\mathbf{d}_A\|_2^2$ and $S_B = \|\mathbf{d}_B\|_2^2$. The reported power discrepancy, symmetric complex residual, and signed cross-correlation are

$$e_P = \left| \frac{S_A}{S_B} - 1 \right|, \quad e_\delta = \frac{\|\mathbf{d}_A - \mathbf{d}_B\|_2}{(S_A S_B)^{1/4}}, \quad r = \frac{\text{Re}(\mathbf{d}_B^\dagger \mathbf{d}_A)}{\sqrt{S_A S_B}}. \quad (\text{A.3})$$

Absolute differences are retained when the signal is undefined; relative metrics are not manufactured by flooring an absent signal. In these density comparisons the saved definition treats $\min(S_A, S_B) \leq 10^{-28}$ as undefined. For plotted shell powers in E, S, and G, $P(k) = L^3 \langle |\hat{\delta}|^2 \rangle$ and no shot-noise subtraction is applied. The separate ZB shell estimator uses periodic CIC assignment, applies the CIC window correction, and subtracts Poisson shot noise as described in Section A.9.

Independent tests exercise the background growth ODE, Einstein–de Sitter integrals, transfer interpolation, normalization quadrature, manufactured spectral forces, CIC scatter/gather, periodic wrapping, and particle migration. The wrapping regression includes exact box endpoints and excursions by multiple box lengths; ownership, IDs and migrated attributes are checked on ranks 1–4. Manufactured spectral-force errors are approximately 2×10^{-15} . These component tests isolate implementation defects before a dynamical comparison is interpreted as evidence about cosmology.

A.2 Linear evolution and refinement

Campaign L uses $L = 200 h^{-1}$ Mpc, $(\Omega_m, \Omega_b, h, n_s) = (0.31, 0.0487, 0.675, 0.965)$ and BBKS transfer. The Gaussian case deliberately sets $\sigma_8 = 0.02$ to remain linear; it is not the later $\sigma_8 = 0.82$ nonlinear

experiment. Most runs evolve from $z = 49$ to 24 . Axis-aligned and oblique $(1, 1, 1)$ perturbations start with density amplitude 10^{-3} ; the late- Λ test evolves $z = 9$ to 0 with amplitude 10^{-4} . Uniform, axial and Gaussian cases cover ranks 1–4, and a separate one-rank/two-thread run checks OpenMP reproducibility.

All uniform configurations retain exactly zero displacement, momentum and force. Axis/oblique growth errors are approximately 0.1561%/0.1578%; late- Λ growth differs from the independent reference by 0.9427%. The Gaussian low- k complex-growth error over 13 selected modes is 0.3332%, and independent reconstruction of the initial Gaussian coefficients gives relative L_2 error 4.804×10^{-12} . The largest recorded mass residual is 1.686×10^{-12} , below the declared 2×10^{-12} criterion.

The 4/8/16-step sequence measures successive-solution temporal orders 1.9881 in position and 1.9954 in momentum. Joint particle/mesh refinement $N_P = N_M = 16, 32, 64$ at 64 steps reduces the growth errors from 5.9551×10^{-3} to 1.5275×10^{-3} and 3.9681×10^{-4} , with reduction factors 3.8985 and 3.8495. This is evidence for the tested linear discretization, not an independent fixed-particle mesh study. The early report contains the measured outcomes but lacks the executable/source hash linkage added in later campaigns; a subsequent rebuilt executable must not be described as cryptographically identified by that report.

A.3 Matched initial conditions against Zarija

Matched physical subset and deterministic probes. Campaigns ZP and ZI compare the original Zarija Lukić generator with IPPL only in their common flat, Gaussian, radiation-free CDM/1LPT subset: $\Omega_\nu = \Omega_r = f_{\text{NL}} = 0$ and $w = -1$. BBKS and a supplied CMBFAST-style table correspond to reference transfer flags 4 and 0. The original reference sources and transfer input were preserved. The scalar probe evaluates the public reference cosmology implementation, independently of particle statistics. Its 4,655 gated comparisons pass; the additional nine excluded DC/legacy-table diagnostics are not passing acceptance tests (one BBKS DC diagnostic and eight table-interval diagnostics).

Across the sampled simulation wavenumbers, transfer discrepancies are below 5×10^{-16} , against a 2×10^{-12} gate. Power-normalization differences are 4.763×10^{-7} for BBKS and 3.255×10^{-5} for the table, against 5×10^{-4} . At $z = 0, 9, 49, 200$, maximum relative differences in D , $dD/d\tau$ and f are 3.334×10^{-7} , 7.255×10^{-7} and 3.921×10^{-7} , respectively, below 2×10^{-5} . The reference’s finite-start growth solution and numerical integration accuracy are distinct from the IPPL growing-mode quadrature.

Ensemble design and convention conversion. The particle experiment fixes $N_P = N_M = 32$, $L = 168.75 h^{-1} \text{Mpc}$ and $(\Omega_m, \Omega_b, h, \sigma_8, n_s) = (0.31, 0.0487, 0.675, 0.82, 0.965)$. Two transfers, starting redshifts 49 and 200, and eight seeds are used: 1234567, 7654321, 104729, 130363, 32452843, 49979687, 67867967 and 86028121. The 64 serial realizations are augmented by 12 IPPL runs on ranks 2, 3 and 4, and eight reference runs on ranks 2 and 4, using the first seed for each transfer/redshift combination. Thus the 84 runs comprise 44 IPPL and 40 reference executions. The uneven three-rank case is an IPPL test, not a claimed reference slab-decomposition test.

Different generators use different random-number algorithms, so identical numeric seeds do not imply equal cross-code phases. Cross-code agreement is therefore assessed statistically. Within-code changes in redshift and transfer use phase-matched comparisons. Rank changes are phase matched only for IPPL; the reference’s MPI runs test statistics because its RNG realization changes with decomposition. ID-derived Lagrangian coordinates, the node/cell-centred lattice offset, the half-cell Fourier phase, transform ordering and velocity units are handled explicitly. For the reference velocity $\mathbf{v}_Z$, the conversion is $\mathbf{p} = a^2 \mathbf{v}_Z / 100$; the physical peculiar velocity is $100 \mathbf{p} / a = a \mathbf{v}_Z$ in km s^{-1} .

The analysis reconstructs displacement and then the $z = 0$ density coefficients from $-i\mathbf{k} \cdot \hat{\psi}_0$; it does not infer the linear spectrum from nonlinear Eulerian CIC density. It excludes DC and every componentwise Nyquist plane and retains one member of each conjugate pair: 14,895 independent complex modes per seed, or 119,160 per eight-seed ensemble. Redshift-reused realizations are not additional independent samples.

Results and statistical meaning. All 1,047 checks pass. Ensemble means of modewise recovered/theoretical power are approximately 0.995737 for IPPL and 0.99816–0.99820 for the reference. Six-standard-error criteria are based on the expected Gaussian statistics; the reference additionally carries a predeclared 5×10^{-4} deterministic normalization allowance. The mean-power half-widths are 0.01738145 and 0.01788145, respectively. Thus these means are consistent with the expected finite-ensemble scatter; they are not a fitted amplitude correction. Figure A.1 distinguishes theoretical one-standard-error bars from the wider acceptance band.

Maximum canonical-momentum relation errors are 1.998×10^{-12} for IPPL and 3.921×10^{-7} for Zarija. Longitudinal-displacement residuals remain below 2.111×10^{-13} in both. Same-transfer redshift-scaling errors are 6.263×10^{-13} and 5.138×10^{-8} ; transfer-scaling errors are 3.169×10^{-9} and 1.604×10^{-5} . The 12 IPPL MPI comparisons give maximum componentwise position RMS $6.501 \times 10^{-16} h^{-1}\text{Mpc}$ and componentwise momentum RMS 1.339×10^{-17} in canonical units (Figure A.2).

An explicit legacy exclusion. The original table’s first weighted transfer entry, about 1.9965×10^7 , is left unnormalized by the reference. Independent integration of this narrow, malformed low- k interval contributes 1.6618 times the legacy routine’s reported total raw variance, revealing that its quadrature misses the discontinuity. Agreement on the simulation band therefore does not validate that interval. A separately normalized copy diagnoses the issue without modifying the original input. Different RNGs, Nyquist handling, this interval, and untested reference physics are exclusions from equivalence; the experiment does not establish feature parity with the whole Zarija code or identical production input formats.

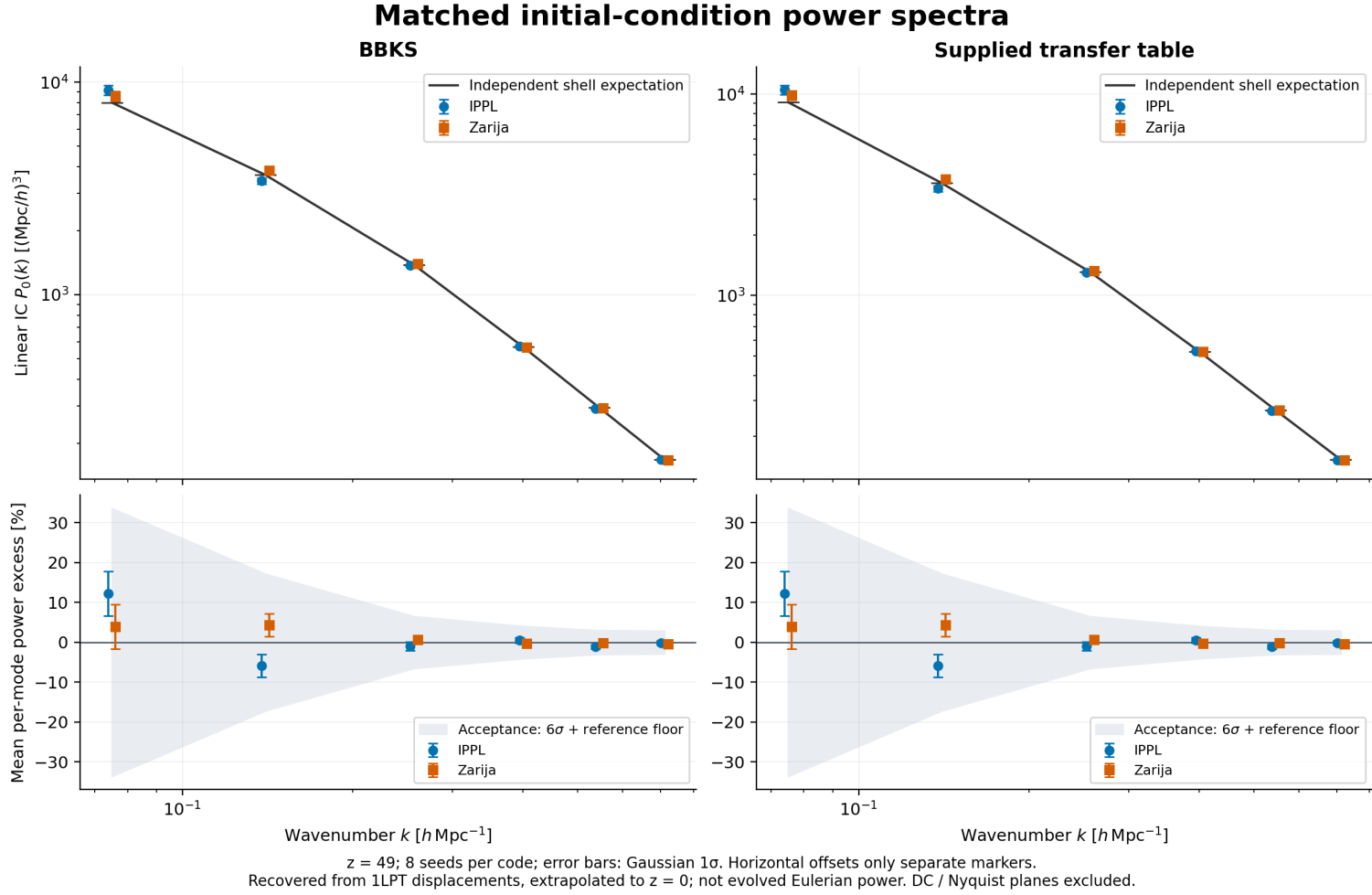


Figure A.1: Recovered $z = 0$ Lagrangian power from eight independent realizations per code at $z_i = 49$, for BBKS and tabulated transfers. Lower panels show ensemble means of modewise power ratios, not ratios of shell means. Error bars are theoretical Gaussian one-standard-error uncertainties; the shaded band is the saved six-standard-error reference criterion including its normalization allowance. DC and componentwise Nyquist planes are excluded; the highest shells have incomplete angular coverage. Source: ZI/ZP.

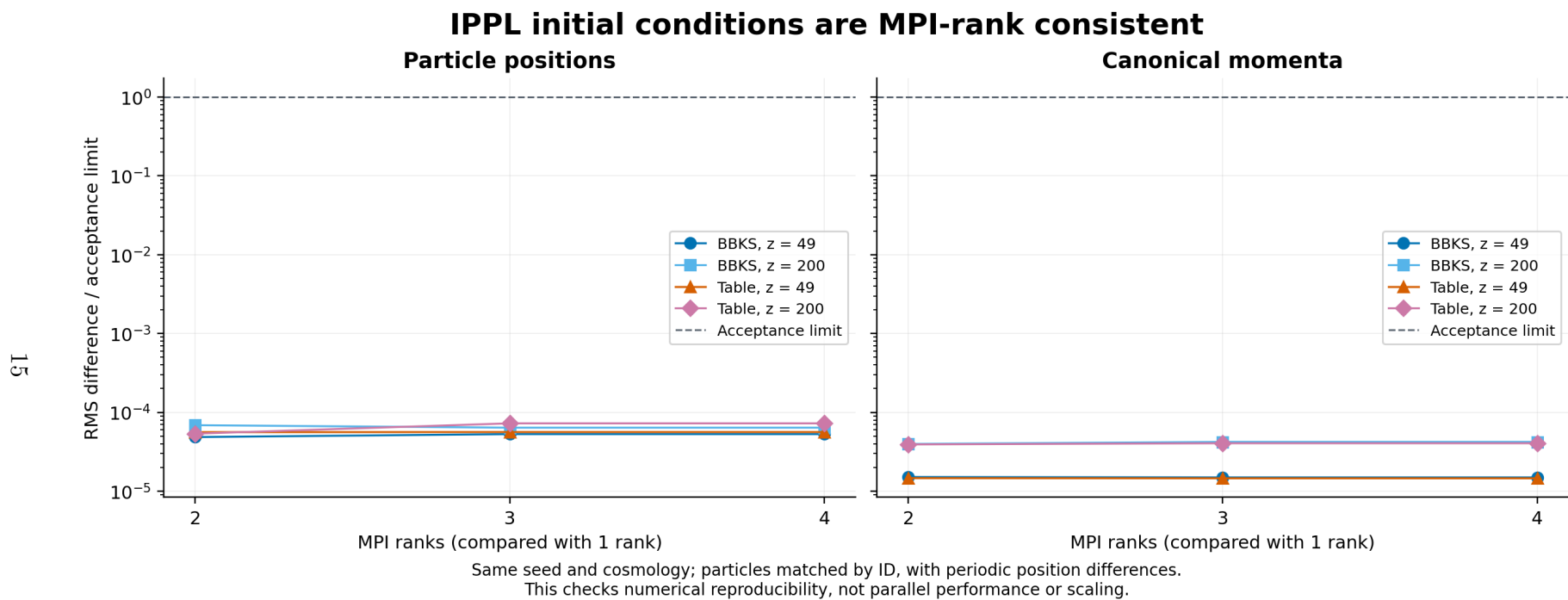


Figure A.2: ID-matched IPPL initial-position and canonical- momentum componentwise RMS differences on ranks 2–4 relative to one rank, normalized by the declared limits. Both transfers and initialization redshifts are included. This is a decomposition-reproducibility test, not a strong-scaling, GPU or throughput measurement. Source: ZI.

A.4 Frozen-force comparison with native FastPM

The frozen experiment isolates scatter, the mesh force, and gather without advancing particles. Reference FastPM [7] is pinned at commit `15b6c4fd7502a81d99dd13f54fcc9cfa44be1331`; the tracked upstream sources are unchanged. Eight fixtures on ranks 1–4 in both codes give 64 runs. They include a uniform state, a strictly common-band mode, axis and oblique deformations, subcell shifts, shuffled/wrapped jitter, a dense cluster, a second mesh size, and an Einstein–de Sitter case. Most use $N = 16$, $L = 168.75 h^{-1}\text{Mpc}$ and $\Omega_m = 0.31$; the matrix also contains $N = 32$ and $\Omega_m = 1$ cases.

The compared force is that of Eq. (3), with CIC scatter and gather and no window deconvolution. Native FastPM uses its NAIVE spectral kernel without softening. The node-mesh origin receives the explicit shift $\mathbf{x}_F = \text{wrap}(\mathbf{x}_I - L/(2N_M))$ in each coordinate. Its exported native acceleration is multiplied by $-3\Omega_m/2$ to obtain the common force convention; this is a documented unit/sign conversion, not a fitted normalization.

The full native discrete operators are *not identical*. IPPL zeros the differentiated component’s Nyquist plane, whereas native FastPM zeros only the eight self-conjugate corners and retains its native real-to-complex behavior. FastPM FFT, mesh and CIC operations are double precision, but its wavevector tables and particle accelerations use float32. Independent NumPy constructions of CIC deposition, each complete native spectral operator and reciprocal gather test each code against its own oracle. The expected difference between these operators is then compared with the actual residual field. A diagnostic common-band projection is never inserted into either production solve.

All 790 checks pass. Maximum relative RMS particle-force errors against the respective independent oracles are 6.741×10^{-14} for IPPL and 3.205×10^{-8} for FastPM; mesh-force errors are 5.672×10^{-14} and 6.474×10^{-14} . The specially constructed common-band fixture has raw particle-force agreement within 4.142×10^{-8} over ranks 1–4. In contrast, raw oblique and wrapped-jitter differences reach 2.467% and 11.714% and are predicted by the preserved Nyquist operators. These are not hidden by a looser raw-force gate. Uniform relative-force errors are undefined when both forces are zero. Figure A.3 reports residual fields, not a difference between the separate force magnitudes.

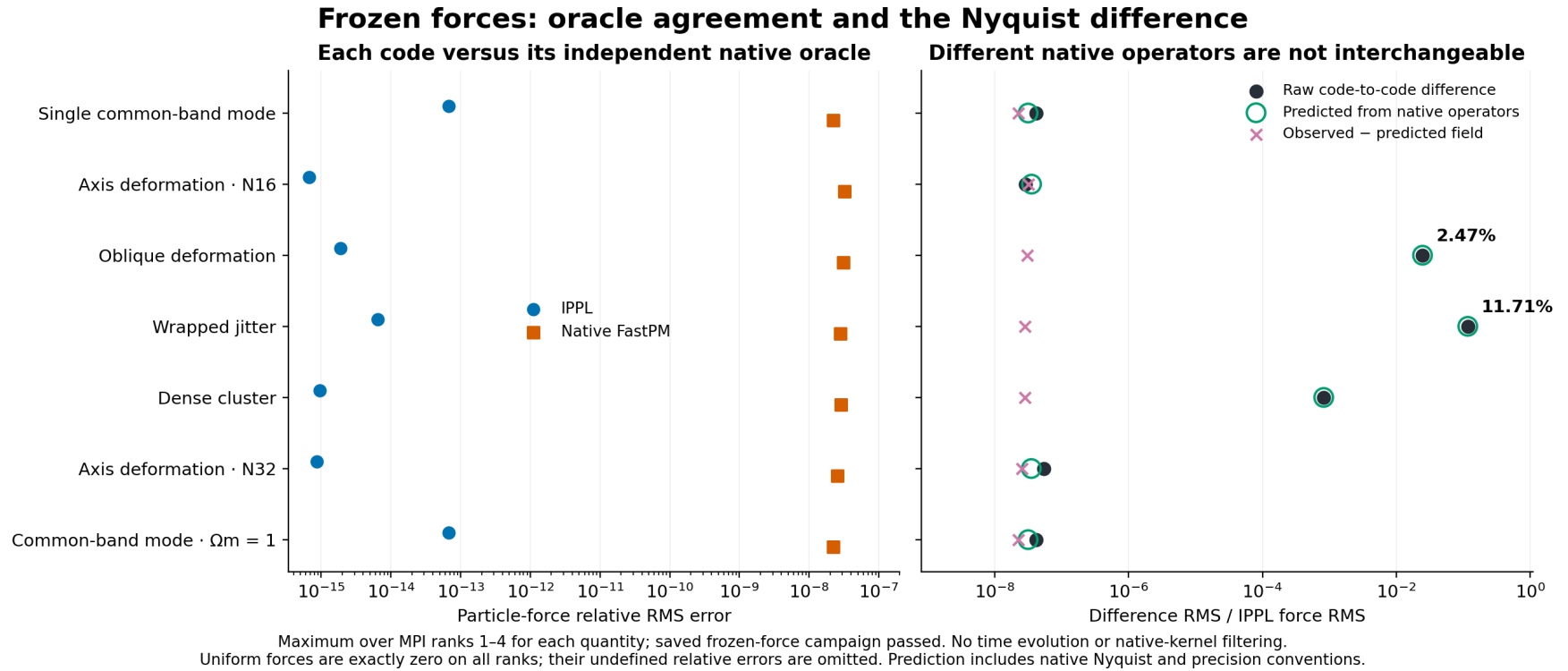


Figure A.3: Frozen operators on ranks 1–4. Each native code is compared with its own independent CIC/spectral/gather oracle; raw and predicted cross-code differences retain the different Nyquist conventions. Values are maxima over tested ranks. Exactly zero uniform-force cases are omitted from relative logarithmic plots, not assigned a positive floor. Passing this test does not make arbitrary unfiltered native forces identical. Source: F.

A.5 Analytical pancake before shell crossing

An independent planar growing-mode solution tests actual evolution, not just the force at a prescribed state. Before crossing,

$$\mathbf{x} = \mathbf{q} - A(a) \sin(\mathbf{k} \cdot \mathbf{q}) \frac{\mathbf{k}}{k^2}, \quad \mathbf{p} = a^2 E(a) f(a) (\mathbf{x} - \mathbf{q}), \quad (\text{A.4})$$

where $A(a)$ follows independently evaluated growth. The Jacobian along the normal is $J = 1 - A \cos(\mathbf{k} \cdot \mathbf{q})$, and $\rho/\bar{\rho} = J^{-1}$ while the map is single valued. Final amplitudes 0.5 and 0.8 give peak density contrasts 1 and 4; the analytic oracle is never continued past crossing.

Campaign P evolves $z = 49$ to 9 and varies coupled particle/mesh sizes 16/32/64, timestep counts 16/32/64/128, axis/oblique orientation and ranks 1–4. Spatial errors compare with continuum trajectories at 128 steps; temporal errors compare successive numerical solutions at fixed mesh. At $N = 64$ and $A_f = 0.8$, displacement and momentum relative RMS errors are 0.413790% and 0.735828%. Spatial orders span 1.4337–1.7603 and temporal orders 1.7994–2.0183. These results pass their engineering trajectory budgets but do not establish second-order spatial accuracy (Figure A.4).

One of 364 checks fails: the mass diagnostic reaches $2.07200923 \times 10^{-12}$ against 2×10^{-12} at step 89, $a = 0.0612396326$, for expected total unit mass 262,144. The absolute reported discrepancy is 5.43165×10^{-7} . Independent endpoint CIC redepositions summed with `math.fsum` recover exactly 262,144, but the worst intermediate state was not saved. Summation roundoff is therefore a plausible explanation, not a proved resolution of that failed gate. Neither a summation-error bound nor the user’s acceptance of a practical engineering baseline changes the original failed status.

Local gradients are a separate limitation. At $A_f = 0.8$, sampled minimum Jacobians change $0.207978 \rightarrow 0.181079 \rightarrow 0.172239$ on 16/32/64, rather than converging monotonically to the continuum value 0.2. Maximum errors against the identically finite-differenced analytical map remain about 0.19. Figure A.5 shows why shrinking support of a trajectory defect can improve a global RMS while leaving local density unresolved. Grid/lattice locking is a possible explanation, not an established cause.

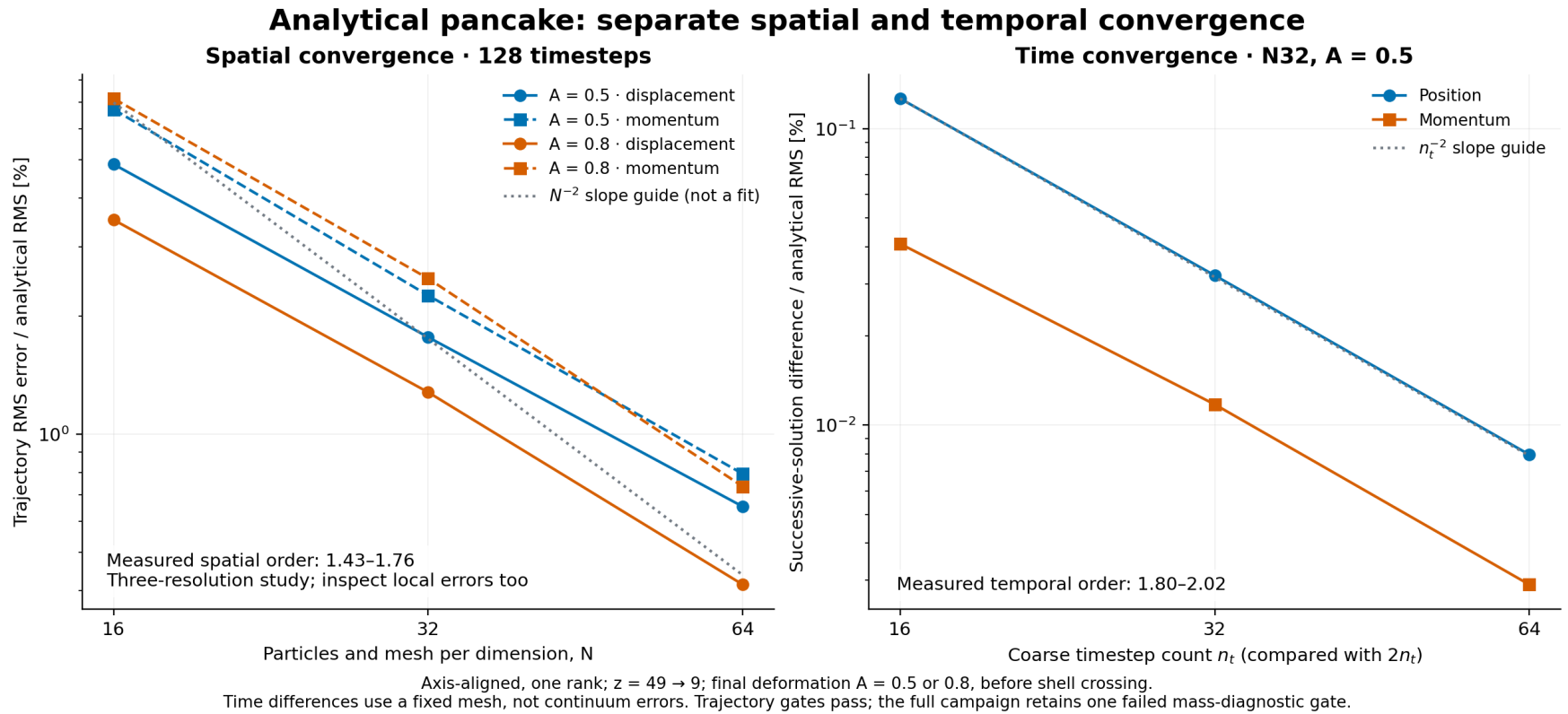


Figure A.4: Analytical pre-crossing pancake: global continuum trajectory errors under joint particle/mesh refinement and successive-solution differences under fixed-mesh temporal refinement. Position and momentum use their respective analytical RMS scales. Slope guides are not fitted results. The original mass-sum failure is retained; passing global trajectory tests does not establish local-density accuracy. Source: P.

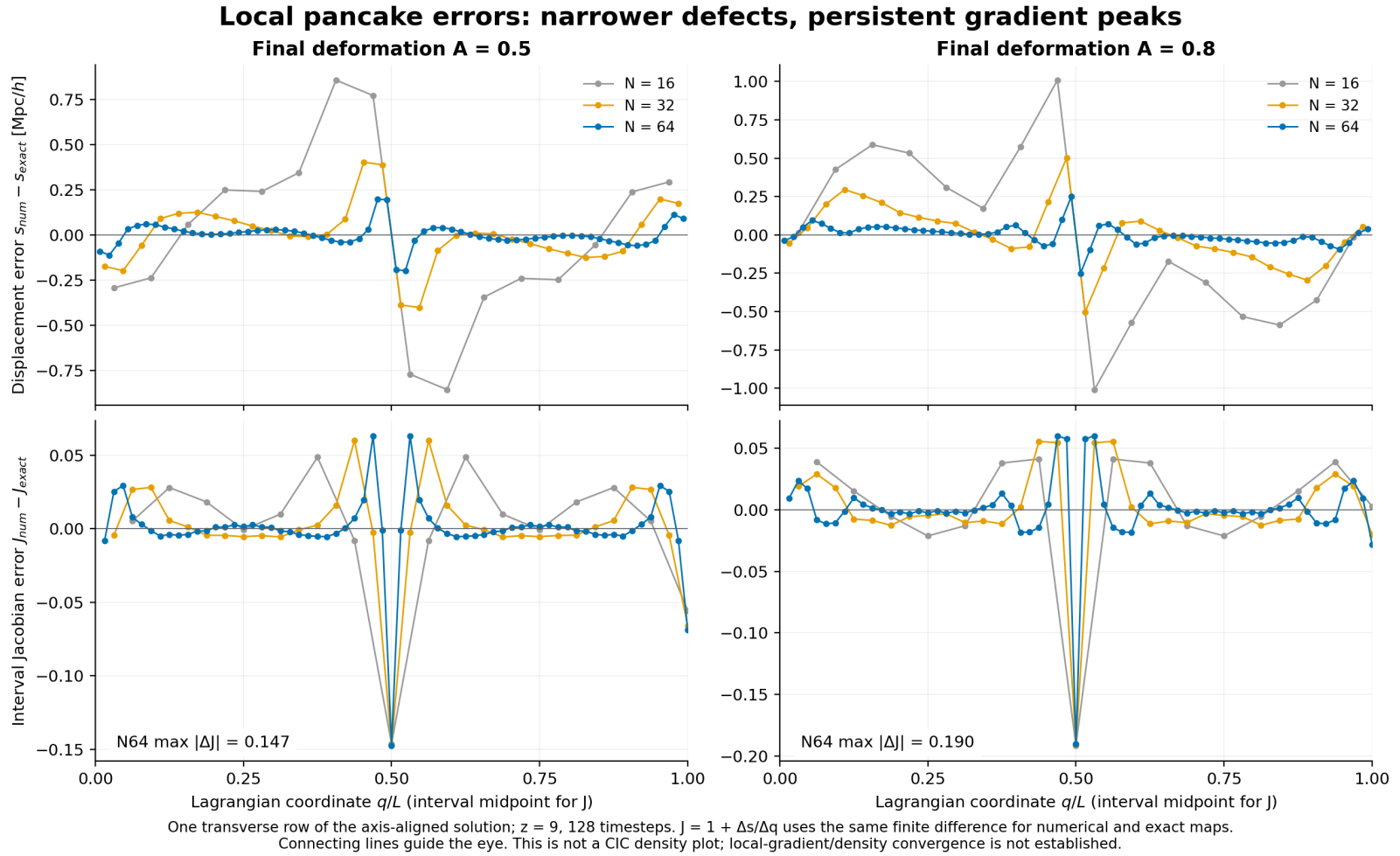


Figure A.5: Local displacement and interval-Jacobian errors along one transverse Lagrangian row. Numerical and exact maps are finite-differenced identically. These are not Eulerian CIC density curves. The persistent local defects explain why the global trajectory results alone cannot qualify peak density or caustic structure. Source: P and its saved diagnostic audit.

A.6 Matched nonlinear evolution and the targeted time refinement

Campaign E advances identical imported positions and canonical momenta with the production IPPL force/KDK methods and native `fastpm_solver_evolve` in `FASTPM_FORCE_PM` mode. It does not use the modified FastPM stepping scheme or COLA. Initial momenta are rounded once to float32 and those same values are supplied to both codes; native momentum/acceleration remain float32 and positions/FFTs remain double. Origin shifts are reversed when actual native states are exported. Logarithmic full-step endpoints, geometric kick midpoints and synchronized native outputs are compared without snapshot interpolation. Actual native kick/drift factors agree with independent quadrature within 6.740×10^{-14} , below the 10^{-8} gate.

The 32-run matrix fixes $N_P = 32$, $L = 168.75 h^{-1} \text{Mpc}$, $\Omega_m = 0.31$ and $a = 0.02 \rightarrow 0.2$. A planar fixture has extrapolated final linear amplitude 1.5 and must exhibit actual sheet folding; a second fixture contains nine coupled three-dimensional growing modes with final linear density RMS 1. The latter is a deterministic nonlinear test, not a σ_8 -normalized Gaussian realization. Meshes 16/32/64, 64/128/256 steps and ranks 1–4 are tested, with nine synchronized checkpoints per run. Unequal particle/mesh counts use the explicit cell-mass normalization N_M^3/N_P^3 before subtracting unity.

The accepted density band consists of 128 unique conjugate pairs with $0 < |\mathbf{n}| \leq 4$, or four active harmonics for the planar fixture, measured with Eq. (A.2). Aggregate power/complex budgets are 5/2/1% for $N_M = 16/32/64$, with minimum correlations 0.995/0.999/0.9995. Planar cross-code position and momentum budgets are 10^{-3} cells and 10^{-3} relative RMS. Raw three-dimensional trajectories and individual shells are characterized rather than gated as identical-operator solutions.

All imported-state, paired-code, rank, native-factor, mean-momentum and folding checks pass. Planar cross-code differences stay below 1.719×10^{-6} cells and 3.821×10^{-7} relative momentum. Coupled-3D aggregate power/complex differences reach only 0.24245%/0.33780%, with $r \geq 0.999994495$. Raw 3D particle differences nevertheless reach 0.03205 cells and 1.0072% relative momentum. Figures A.6 and A.7 illustrate this matched discrete comparison, not an analytical solution after crossing.

The two failures in E are final 128-to-256-step momentum differences: 0.276343% for IPPL and 0.275555% for native FastPM, exceeding the unchanged 0.2% criterion despite reduction factors near four. The separate two-run ER extension adds 512 steps only at $N_P = N_M = 32$ on one rank. Its 256-to-512 differences are 0.0696213%/0.0694281% in momentum and 0.000449258/0.000448378 cells in position; all 99 checks pass, with final observed orders 1.9725–2.0083 for the assessed 3D quantities. Below the predeclared analysis floors no order is inferred. ER does not relabel E as all-pass and supplies no unperformed 512-step MPI comparison.

Fixed-particle mesh sensitivity is not continuum convergence: the pancake power changes increase from 0.5367% on 16-to-32 meshes to 3.6296% on 32-to-64. The maximum IPPL CIC mass residual in E is 1.219×10^{-12} ; that smaller residual does not overturn P’s distinct failed mass check. Native unit-particle counts and IPPL deposited-CIC sums are different diagnostics and are never treated as interchangeable measures.

Synthetic planar collapse · $N_p = 32^3$, $N_m = 32$, 256 steps, 1 MPI rank

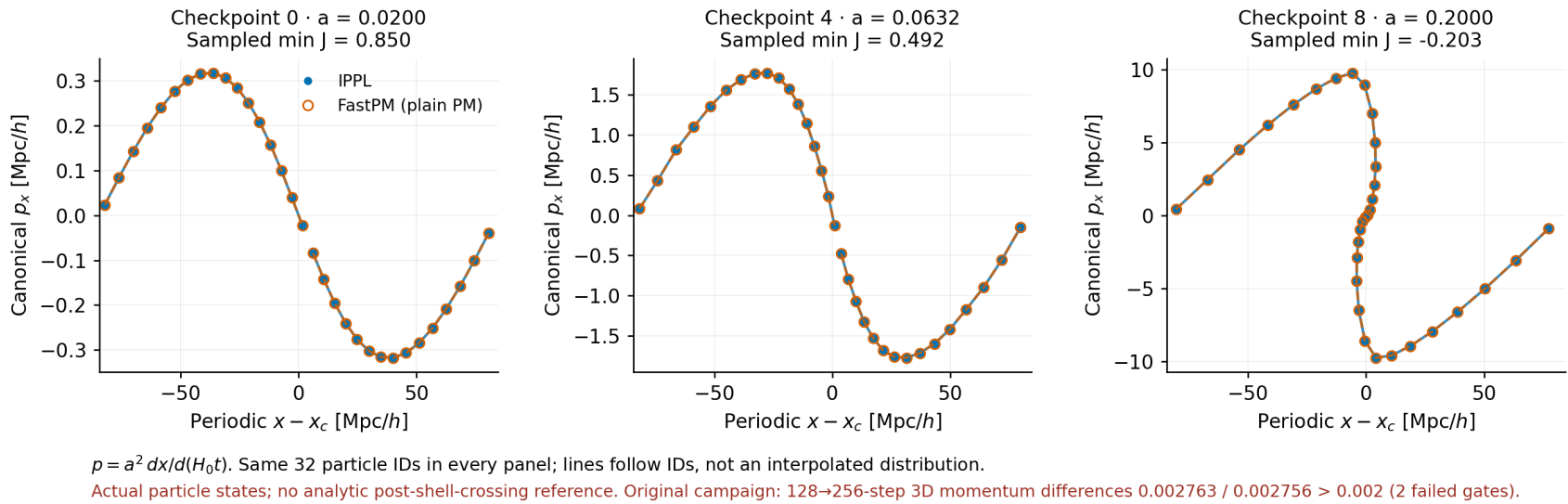


Figure A.6: Matched particle-sheet phase portraits at three saved epochs, including the folded final sheet. Curves connect the same 32 ID-matched particles in one transverse Lagrangian row, with breaks at periodic edges; they are not an interpolated distribution. The pre-crossing analytical result is not an oracle for the final folded state. Source: original 32-run campaign E.

Direct particle-density modes · fixed particles, 256 steps, 1 MPI rank

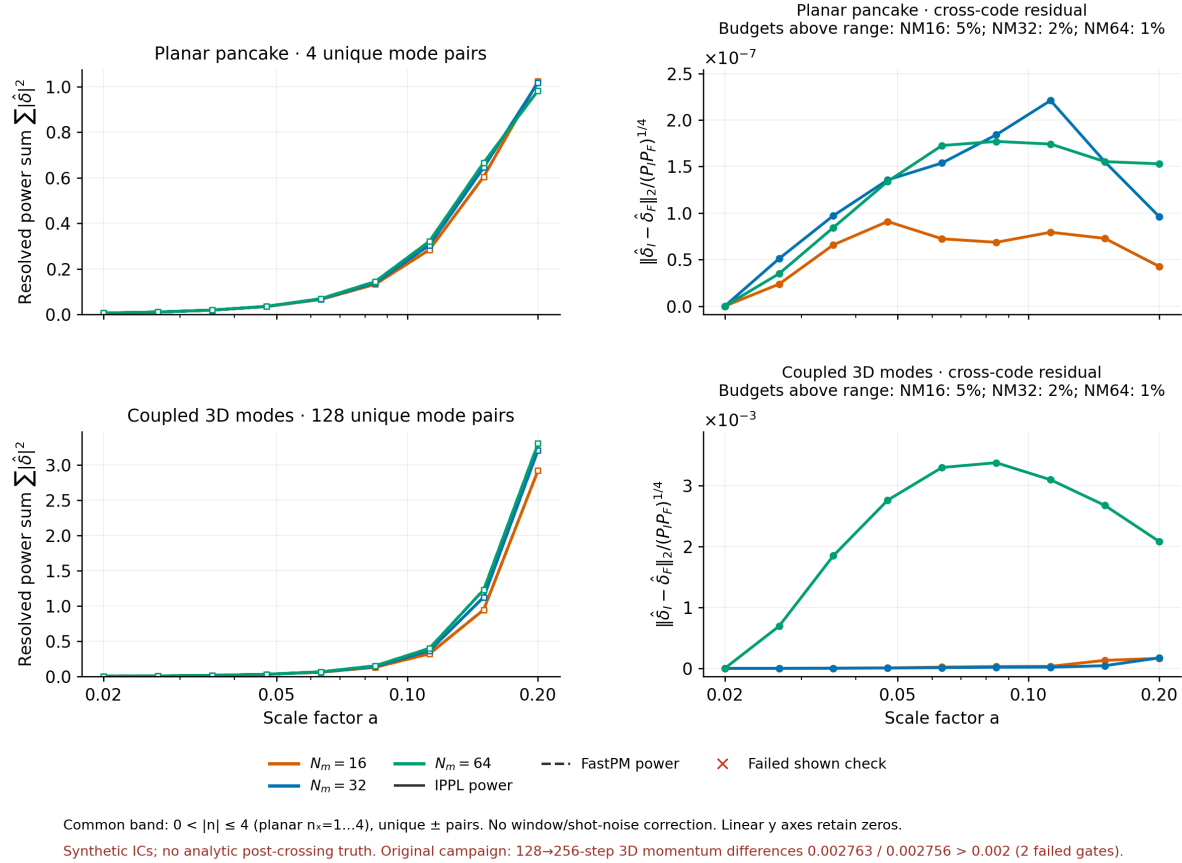


Figure A.7: Resolved-density evolution from exact particle sums, with $N_P = 32$, meshes 16/32/64, 256 steps and one rank. The 3D comparison uses 128 unique low-mode pairs; the pancake uses four active harmonics. The figure retains E's two failed timestep checks, which are not failures of the plotted cross-code density comparison. The separate 512-step follow-up ER is described in the text, not substituted into this figure.

A.7 Crossed spatial resolution and translation controls

Campaign S removes an ambiguity of earlier tests by varying particle sampling and force-mesh size independently. For each planar/coupled-3D fixture, it crosses $N_P = 32, 64$ with $N_M = 32, 64$ at 1,024 steps, adds rigid translations at $N_P = 64$ on both meshes, and adds 512/2,048-step cases at $N_P = N_M = 64$. The coupled-3D finest standard case is also repeated on three ranks. Seventeen paired cases give 34 runs, with the same L , Ω_m and $a = 0.02 \rightarrow 0.2$ as E and nine saved epochs per run. The fixed physical translation is $\Delta = (0.37, 0.23, 0.41)L/64$; its known Fourier phase is removed before comparing coefficients.

Direct modes with $0 < |\mathbf{n}| \leq 4$ are separated into three shells: $0 < |\mathbf{n}| < 1.5$, $1.5 \leq |\mathbf{n}| < 2.5$, and $2.5 \leq |\mathbf{n}| \leq 4$. A bookkeeping edge at 4.5 does not extend the measured band beyond 4. For campaign G only, fixed 128^3 interlaced PCS extraction, with its assignment-window correction, characterizes higher modes through 12. Its low-mode coefficients must independently agree with the direct sums within $\max(10^{-12}, 10^{-3}\|\mathbf{d}_{\text{direct}}\|_2)$. PCS is an analysis tool; the simulation forces remain CIC.

Table A.2: Predeclared shell-wise controls in S and G. Percentages denote e_P or e_δ in Eq. (A.3), not errors against truth. All applicable saved epochs and both codes enter the controls.

Control	e_P limit	e_δ limit	Minimum r
Paired codes, $N_M = 32$	2%	2%	0.999
Paired codes, $N_M = 64$	1%	1%	0.9995
Particle or mesh refinement	5%	5%	0.999
Dephased translation (S)	1%	2%	0.9998
Starting redshift (G, final epoch)	2%	3%	0.999
Finest timestep shell difference	—	0.2%	—

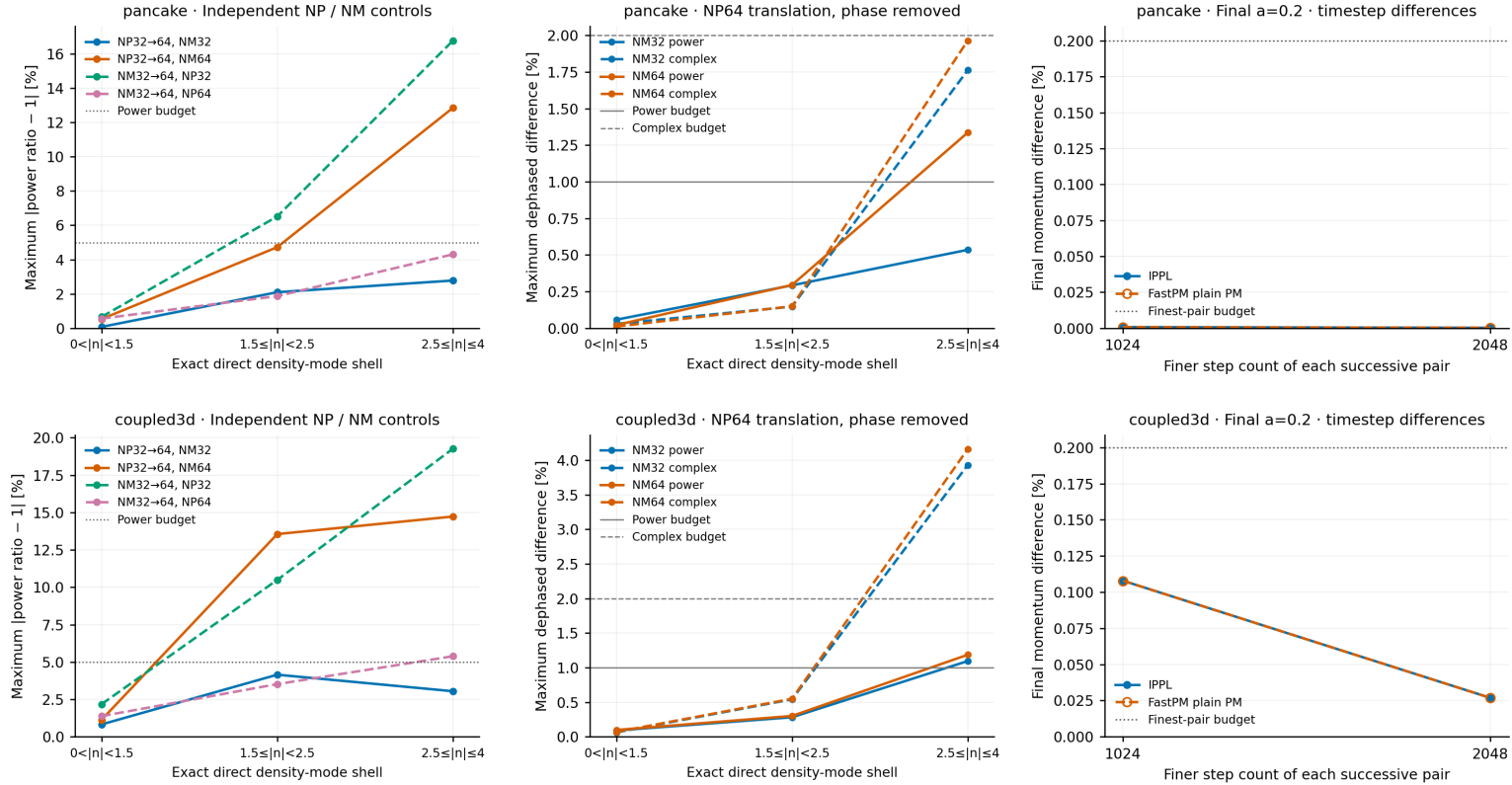
Qualification requires a contiguous prefix of shells for which *every* applicable control passes. Failed integrity, extraction, rank or global-time controls veto qualification. Passing an aggregate low-mode norm or choosing a favorable final epoch is not an alternative to this rule.

Of 2,213 checks, 130 fail: 56 particle-resolution, 51 mesh-resolution, 20 translation and three paired-code shell checks. Both fixtures qualify only the first shell, $0 < |\mathbf{n}| < 1.5$, over the declared matrix. For the 3D fixture the largest measured mode in that shell is $\sqrt{2}$; for the planar fixture it is the fundamental mode. All required global, timestep and rank controls pass.

The maximum power sensitivity is 16.77% for the pancake and 19.27% for coupled 3D. The finest-mesh translated pancake reaches 1.336% power difference versus 1%; coupled-3D translation reaches 4.160% complex difference versus 2%. The three cross-code failures occur only for the undersampled $N_P = 32$, $N_M = 64$ coupled-3D third shell at intermediate epochs: complex differences of approximately 1.11–1.39% exceed 1%, while the power and correlation conditions pass. Close matched $64^3/64^3$ agreement does not remove these failures from the crossed matrix.

Final 1,024-to-2,048-step coupled-3D momentum differences are approximately 2.683×10^{-4} relative, with near-second-order late-time reduction. Some pancake differences are below the predeclared analysis floor, and no order is inferred for them. Three-rank differences stay below 5.9×10^{-15} cells/ 5.1×10^{-15} relative momentum for IPPL and 1.16×10^{-7} cells/ 8.63×10^{-8} for native PM. Four separate IPPL pancake runs also exceed the historical 2×10^{-12} mass diagnostic, with maximum 2.951×10^{-12} . These explicitly retained historical flags are not recast as passing tests or added to the new shell-gate count. Figure A.8 shows why code agreement alone would have missed the particle-sampling and grid-alignment sensitivities.

Spatial robustness controls · direct particle density modes only



Recorded spatial stage: 130 failed checks retained. Prefixes — **pancake**: 1/3 low shells, **coupled3d**: 1/3 low shells.
 Translation $\Delta = (0.37, 0.23, 0.41)L/64$. Control maxima span saved epochs and both codes; exact $|n| \leq 4$ only. Differences are not truth errors.

Figure A.8: Independent particle and force-mesh controls on the local CPU platform. Rows show planar and coupled-3D fixtures. Power/translation panels are worst-case envelopes over both codes and nine saved epochs, not ensemble uncertainties or continuum errors. Translation phases are removed analytically. Temporal panels compare 512/1,024/2,048 steps at $N_P = N_M = 64$. Only the first exact shell meets every declared control; all 130 failures remain recorded. Source: S.

A.8 Common-phase Gaussian evolution on Merlin

Campaign G is a fresh Linux/OpenMP experiment, not a resumed macOS journal. It uses one mode-keyed Gaussian BBKS realization (seed 20261003) with common physical phases at both particle resolutions, initial band $0 < |\mathbf{n}| \leq 12$, $L = 168.75 h^{-1}\text{Mpc}$, and $(\Omega_m, \Omega_b, h, n_s, \sigma_8) = (0.31, 0.0487, 0.675, 0.965, 0.82)$. The BBKS choice has no baryonic acoustic transfer features; the baryon parameter does not make this a full baryon-physics experiment. σ_8 normalizes the continuous spectrum, not the measured finite box or truncated realization. Both codes receive the identical imported, rounded-once canonical momenta.

Nine paired cases comprise the four crossed $N_P, N_M = 32, 64$ combinations at $z_i = 49$, 2,048 steps; finest-grid 1,024/4,096-step controls at $z_i = 49$; finest-grid 2,048/4,096-step controls at $z_i = 99$; and the standard finest case on four ranks. All 18 runs end at $a = 1$, saving nine synchronized epochs. Starting-redshift comparisons use only the final epoch at 4,096 steps: intermediate indices denote different scale factors for different starts. No interpolation is used.

The build uses GCC 14.3, OpenMPI 5.0.10, Kokkos 5.2.0, heFFTe 2.4.1 and the pinned native reference. Metadata records OpenMP/Host execution and actual MPI size: at most four ranks/workers, one thread per nonlinear rank, no GPUs. Login-host timings are not performance results. All 21 cosmology CTests pass, including ranks 1–4; an eight-run pipeline smoke passes 364 checks. Neither substitutes for the full scientific matrix.

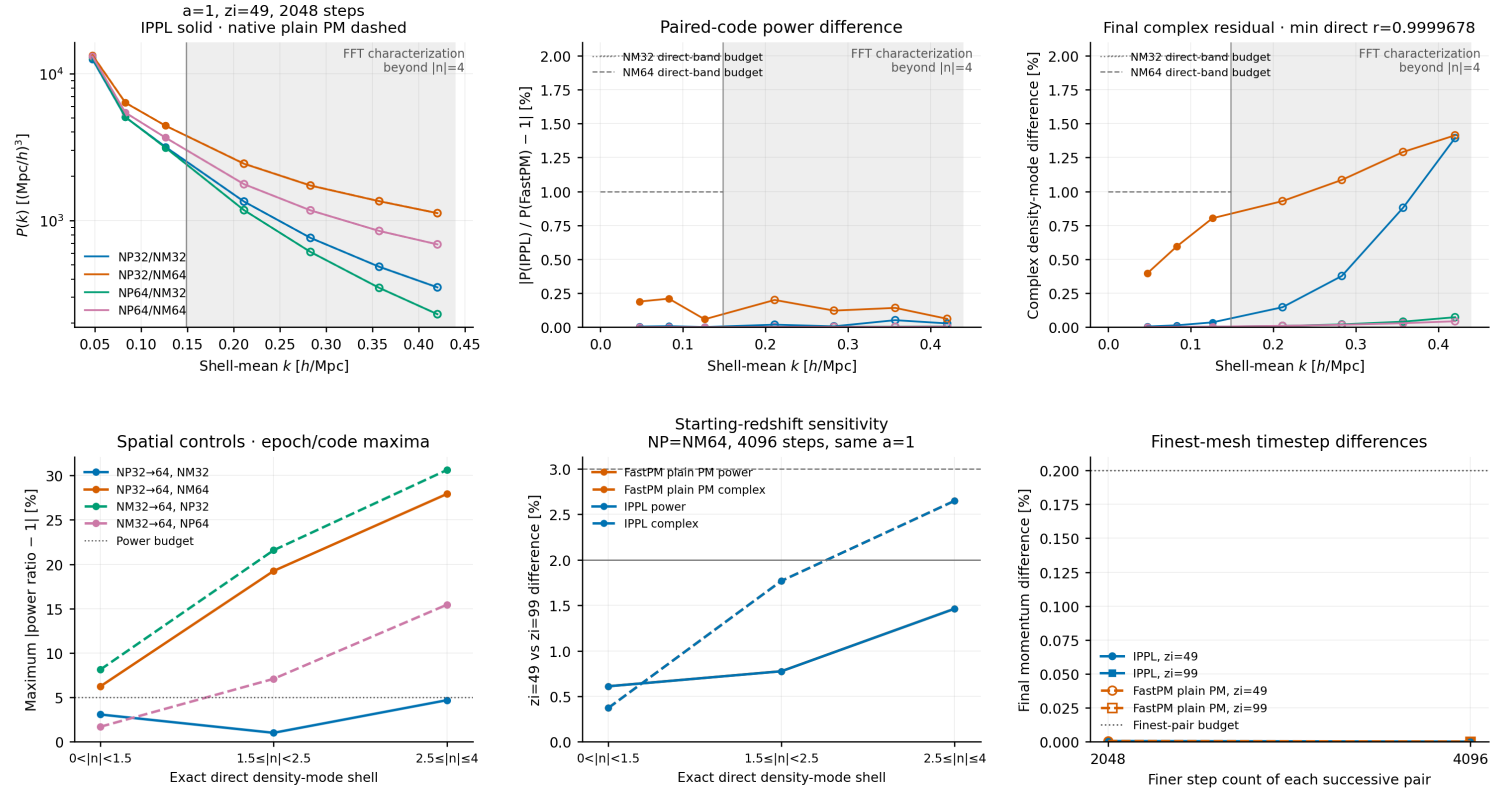
All 1,355 checks are present: 1,239 pass and 116 fail, equally divided between particle and mesh resolution. Every integrity, measurement, paired-code, timestep, start-redshift and rank check passes, but *none of the three shells qualifies across the complete resolution matrix*. Lowest-shell power sensitivities reach 6.246% (particles) and 8.137% (mesh) at intermediate epochs, exceeding 5%; its complex and correlation criteria pass. Higher-shell power changes reach 27.91% and 30.61%, respectively, in both implementations. No retrospective subset or relaxed tolerance removes them.

Table A.3: Gaussian CPU results in G. Paired-code values concern the exact direct band $0 < |\mathbf{n}| \leq 4$. Finest-case values are at $a = 1$, $N_P = N_M = 64$, $z_i = 49$, 2,048 steps. Temporal/rank differences are not truth errors.

Observable	Result
Largest paired-code power / complex difference	0.211604% / 0.802740%
Minimum paired-code correlation	0.999967822
Finest-case power / complex difference	0.00120759% / 0.00362866%
Finest-case minimum correlation	0.99999999342
Starting- z power / complex sensitivity	1.463041% / 2.649273%
Largest final 2,048-to-4,096 position difference	3.970×10^{-6} cells
Largest final 2,048-to-4,096 momentum difference	2.448×10^{-6} relative
Largest shell temporal complex difference	1.303×10^{-6}
Rank 1 versus 4, IPPL position / momentum	3.14×10^{-15} cells / 1.26×10^{-15} relative
Rank 1 versus 4, native PM position / momentum	2.33×10^{-8} cells / 1.85×10^{-8} relative

Final $z_i = 49$ position orders are 2.0003 for IPPL and 1.7405 for native PM. The corresponding final momentum/density differences are below the predeclared analysis floor; this does not demonstrate machine-precision saturation. The two $z_i = 99$ step counts measure a refinement difference, not an order. The largest Gaussian IPPL mass residual is 5.018×10^{-14} , without changing any earlier mass failure. The practical conclusion of Figure A.9 is strong matched-discretization agreement but insufficient particle/mesh robustness in the tested 32/64 matrix.

Common-phase band-limited Gaussian Λ CDM · one realization, not a precision-cosmology claim



Recorded gaussian stage: 116 failed checks retained. Prefixes — gaussian: 0/3 low shells.

IC band $|n| \leq 12$. Filled: exact direct $|n| \leq 4$; open/shaded: FFT characterization only. No shot-noise subtraction. Differences are not truth errors.

Figure A.9: Merlin OpenMP evolution of one common band-limited Gaussian BBKS realization. Top: final spectra and paired-code residuals for the crossed 32/64 grids at $z_i = 49, 2,048$ steps. Filled symbols use exact modes $|n| \leq 4$; open symbols and the higher shaded band are PCS characterization only. Bottom: worst-epoch spatial sensitivities, starting-redshift changes at 4,096 steps, and time refinement. All 116 resolution failures remain visible, leaving zero qualified shells. Source: G.

A.9 Broadband Zel’dovich evolution against native FastPM

Campaign ZB follows the low-band frozen-force and nonlinear checks with a matched, broadband particle evolution. One Gaussian realization (seed 20261003) supplies common 1LPT initial conditions at $z_i = 99$ and 49. The spherical Fourier band is $0 < |\mathbf{n}| \leq 48$, or 75% of the particle Nyquist frequency for $N_P = 128$. The periodic box is $L = 168.75 h^{-1}$ Mpc with a 128^3 particle load and a 128^3 CIC force mesh. The recorded background uses $(\Omega_m, h, n_s, \sigma_8) = (0.31, 0.675, 0.965, 0.82)$ and a pure BBKS transfer. Although a baryon density is recorded in the input metadata, BBKS here has no baryon-acoustic features; the experiment does not model radiation, neutrinos, gas, or baryonic effects.

Both solvers import the same once-rounded float32 particle momenta, without growth rescaling. IPPL uses the production particle–mesh kick–drift–kick adapter; the pinned native FastPM run uses ordinary plain-PM evolution, not FastPM’s modified fast-force stepping or COLA. Thus ZB is a cross-code comparison of a closely matched PM model, not a comparison with GADGET-2 or an independent short-range gravity method. Each case advances to $a = 1$ with 24 uniform-in- $\ln a$ intervals (2,400 steps from $z_i = 99$ and 2,040 from $z_i = 49$); the two starts have nearly equal step size in $\ln a$. The change in start epoch and step count is a sensitivity comparison, not a convergence order test.

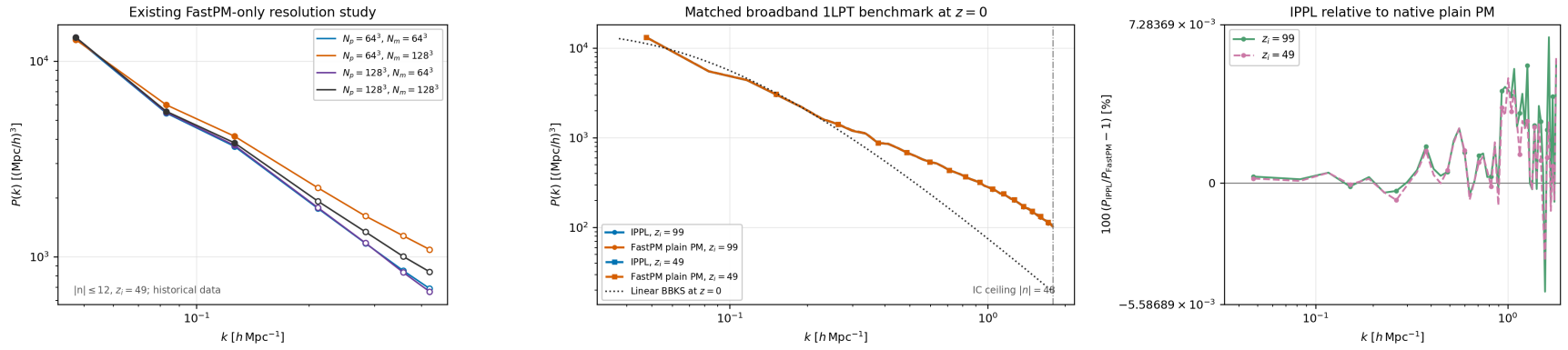
Four launches ran sequentially on the Merlin login CPU using eight single-threaded MPI ranks. All completed under the fixed 1,200-second per-launch limit. Each of the four runs saved 25 checkpoints on eight ranks; all 800 archived rank shards, output metadata, source/executable provenance, and input hashes verified. The imported initial positions round-trip within 7.3×10^{-16} force-mesh cells and momenta match exactly. The largest recorded IPPL checkpoint mass residual is 7.11×10^{-15} . These are integrity and operational checks; no threshold was imposed on the final cross-code scientific differences.

At each saved epoch, density was assigned periodically by CIC to the 128^3 analysis mesh, the CIC power window was corrected, and Poisson shot noise was subtracted. Shells include modes through $|\mathbf{n}| = 48$. Table A.4 reports final particle-phase-space differences and the largest absolute binwise $P(k)$ ratio difference among the 48 shells where both shot-noise-subtracted powers are positive.

Table A.4: Final $z = 0$ cross-code comparison in ZB. Momentum differences are relative to native FastPM. The power maximum is over 48 positive shot-noise-subtracted shells through the initial-condition mode ceiling.

z_i	Position RMS [Δx]	Momentum RMS [relative]	$\max P_{\text{IPPL}}/P_{\text{FastPM}} - 1 $	Positive shells
99	6.0085×10^{-3}	1.7052×10^{-3}	6.6987×10^{-5}	48
49	5.1486×10^{-3}	1.6049×10^{-3}	5.7355×10^{-5}	48

The maximum power differences are 0.00670% ($z_i = 99$) and 0.00574% ($z_i = 49$). This close agreement is consistent with the shared realization and closely matched PM force and integration path. It does not establish agreement with a continuum solution: the run has one realization, uses a single particle/mesh resolution, and its high- k bins are characterization near the particle Nyquist, not a qualified accuracy band. The logarithmic power spectrum and ratio are shown in Figure A.10. The left panel is the older cutoff-12 FastPM-only resolution study and is kept separate from the new matched broadband measurements.

Shared-realization Λ CDM particle evolution: IPPL vs FastPM

128³ particles / 128³ CIC mesh; $L = 168.75 \text{ Mpc}/h$; $|n| \leq 48$ shared Gaussian 1LPT IC; $\Omega_m = 0.31$, $\sigma_8 = 0.82$; eight single-threaded ranks. Left: separate older $|n| \leq 12$ data. CIC-window corrected; Poisson shot noise subtracted. High- k bins are characterization, not a convergence claim.

Figure A.10: Matched broadband 1LPT evolution at $z = 0$. Center: periodic-CIC, CIC-window-corrected, shot-noise-subtracted spectra from the shared-realization 128³ runs and the linear BBKS spectrum. Right: binwise IPPL/native plain-PM FastPM power ratio difference for the two starting redshifts. The left panel preserves the historical cutoff-12 FastPM-only resolution study; it is not merged with ZB. The high- k points are characterization, not a convergence claim. Source: ZB.

A.10 Matched initial-redshift-99 evolution with GADGET-2 TreePM

Campaign GT adds an independently implemented gravity solver to the saved ZB $z_i = 99$ comparison. It is a matched-realization comparison, not a random-seed-matched ensemble: IPPL and native FastPM use their common saved Gaussian 1LPT phase-space realization, and GADGET-2 is initialized from the same CSV fixture (SHA-256 recorded in the GT provenance manifest). The GADGET format-1 conversion changes units and stores position and velocity blocks in single precision; its round-trip audit reports position RMS 3.43×10^{-6} force-mesh cells and relative momentum RMS 2.53×10^{-8} . These are small conversion residuals, not exact bitwise equality of the three solver input files.

All three integrations use the same periodic flat-CDM background, box $L = 168.75 h^{-1}$ Mpc, 128^3 particles, and the same saved initial phases at $z_i = 99$. The GADGET-2.0.7 run uses periodic TreePM [14] with PMGRID = 128, a short-range tree force, and a halo softening of $26.367 h^{-1}$ kpc. Its declared controls include `ErrTolIntAccuracy` = 0.025, `MaxRMSDisplacementFac` = 0.2, `MaxSizeTimestep` = 0.025, `ErrTolTheta` = 0.5, and `ErrTolForceAcc` = 0.005. It uses adaptive synchronized timesteps. IPPL and the native FastPM run instead use plain particle-mesh forces on a 128^3 force mesh and the fixed, uniform-in- $\ln a$ schedule described for ZB. Thus the underlying collisionless cosmological model and initial phases are matched, but force resolution, short-range treatment, time stepping, and input quantization are not identical. The cross-code difference cannot be attributed to the TreePM correction alone.

What a Fourier shell means. Here “shell” means a radial bin in Fourier space, not a layer of particles in the simulation box. For integer lattice mode $\mathbf{n}$, the mode wavenumber is $\mathbf{k} = (2\pi/L)\mathbf{n}$. The bin index is $b = \lfloor |\mathbf{n}| + 1/2 \rfloor$, with $b = 1, \dots, 48$; all modes in one bin have similar $|\mathbf{k}|$. The plotted center k_b is the full-lattice-multiplicity-weighted mean $|\mathbf{k}|$ in that bin. The shell power is the correspondingly weighted mean of the CIC-window-corrected mode powers, minus Poisson shot noise $1/\bar{n} = L^3/N_P^3$. Consequently, the horizontal axis is k in $h \text{ Mpc}^{-1}$ and $P(k)$ has units $(\text{Mpc}/h)^3$. The first bin is the largest-scale measurement; larger bin numbers probe progressively smaller scales.

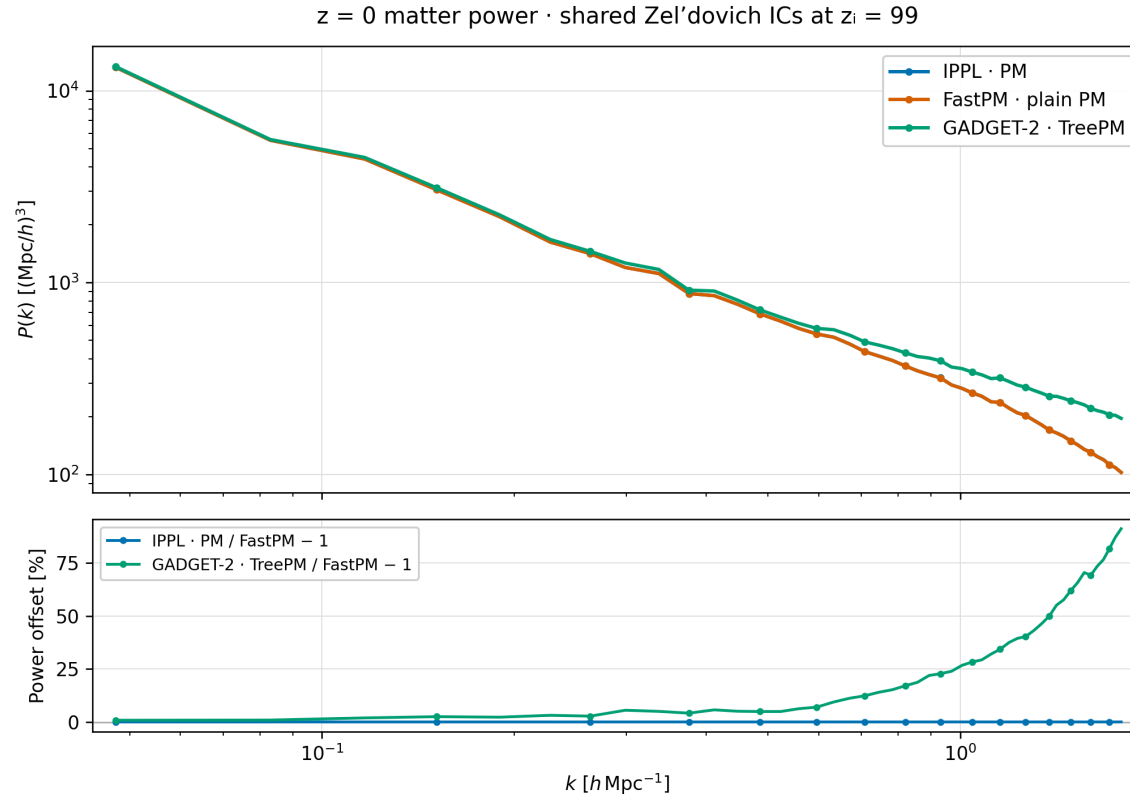
The completed GADGET evolution used eight single-threaded MPI ranks, resumed from a verified eight-rank checkpoint, and exited normally at $a = 1$. The final snapshot contains 2,097,152 dark-matter particles and records $z = 4.4 \times 10^{-16}$, numerically zero. Its measured solver wall time was 2,891.7 seconds. The same external periodic-CIC estimator was applied to all three final particle sets on a 128^3 analysis mesh, including CIC power-window deconvolution and Poisson shot-noise subtraction. All 48 bins have positive subtracted power for each code; they extend through $|\mathbf{n}| = 48$, or 75% of the particle Nyquist frequency. These upper bins are retained as characterization, not as a convergence-qualified science band.

Table A.5: Selected binwise $z = 0$ power differences for the shared $z_i = 99$ realization. Offsets are $100(P_A/P_{\text{FastPM}} - 1)$; the final column uses all 48 positive shot-noise-subtracted bins. No pass/fail threshold was set for GT.

Pair	Bin 1 ($k = 0.0475$) [$h \text{ Mpc}^{-1}$]	Bin 48 ($k = 1.7873$) [$h \text{ Mpc}^{-1}$]	Max. absolute offset over 48 bins [%]
IPPL PM / FastPM plain PM	+0.00029%	+0.00558%	0.00670%
GADGET-2 TreePM / FastPM plain PM	+0.802%	+91.241%	91.241%

The IPPL and FastPM spectra nearly overlap over this measured realization. GADGET-2 is close in the first few bins, then rises above the plain-PM spectrum toward high k ; the lower panel reports these offsets relative to FastPM. This trend is a useful cross-code diagnostic, not evidence that

one curve is the continuum answer: the force models and time integrators differ, there is only one realization and one resolution, and CIC correction does not remove finite-mesh aliasing or nonlinear sampling effects. In particular, the 91% top-bin difference must not be read as a qualified 91% force error.



128³ particles; 128³ CIC mesh; $L = 168.75 \text{ Mpc}/h$; shared Gaussian 1LPT seed 20261003.
 $\Omega_m = 0.31$, $\sigma_8 = 0.82$. Common periodic CIC estimator, window deconvolution, and Poisson shot-noise subtraction.
 GADGET-2 TreePM includes a short-range tree; IPPL/FastPM are plain PM. Modes $|\mathbf{n}| \leq 48$ are characterization, not a convergence claim.

Figure A.11: Final $z = 0$ matter power for the shared $z_i = 99$ Gaussian 1LPT realization. Upper panel: shot-noise- subtracted spectra from IPPL plain PM, native FastPM plain PM, and GADGET-2 TreePM using a common periodic-CIC estimator. Lower panel: binwise power offsets relative to FastPM. A Fourier shell is a radial bin in $|\mathbf{n}|$, not a configuration-space particle shell. The GADGET TreePM short-range force and adaptive synchronized steps differ from the two plain-PM runs. The high- k end is characterization, not a convergence claim. Source: GT.

A.11 Evidence integrity, reproducibility, and remaining claims

Report identifiers below are relative to `build_openmp/demos/cosmology/` in the development worktree, not archival DOIs. The `figures/validation/asset-manifest.json` records copied plots, original manifests, source-report and asset hashes. These are preserved results, not newly executed simulations.

Table A.6: Report identifiers for the completed campaigns. S uses its released spatial snapshot; G uses the separately archived Merlin report.

Key	Authoritative report or exact report snapshot
L	<code>cosmology-validation-na4rs6e0/results.json</code>
ZP	<code>zarija-physics-hd3av78e/results.json</code>
ZI	<code>zarija-validation-f2c9zb42/results.json</code>
F	<code>frozen-force-5_li99qh/results.json</code>
P	<code>pancake-validation-rs1o9glb/results.json</code> ; local-gradient audit: <code>pancake-diagnostics-rs1o9glb/diagnostics.json</code>
E	<code>matched-evolution-5v7u3y98/results.json</code>
ER	<code>evolution-refinement-k2qeaylz/results.json</code>
S	<code>resolution-study-cffjva__/figures-spatial-release/input-report.json.gz</code>
G	<code>merlin-cpu-login-20261004/figures-gaussian-release/input-report.json.gz</code>
ZB	<code>merlin6:/data/user/adelman/cosmology-zeldovich-broadband-20261004/analysis/</code> <code>results.json</code> ; campaign: <code>campaign.json</code> ; plot manifest: <code>plots/manifest.json</code>
GT	<code>merlin6:/data/user/adelman/gadget2-zeldovich-20261004/analysis/</code> <code>gadget2-z99-z0-results.json</code> ; campaign: <code>merlin6:/data/user/adelman/gadget2-zeldovich-20261004/campaign.json</code> ; plot manifest: <code>plots/three-code-z99-z0-v3/manifest.json</code>

The local spatial stage is complete even though its parent macOS journal later stopped safely on a disk guard at 44/52 planned runs. Its ten partial Gaussian runs are not presented as a completed Gaussian experiment. G reran all 18 Gaussian cases on Linux with fresh compiler, executable and absolute-path provenance. A separate Mac/Linux IC comparison found identical canonical momenta but position RMS differences about $1.9 \times 10^{-15} h^{-1} \text{Mpc}$; the CSV bytes are not identical. This is an IC portability observation, not a nonlinear cross-platform trajectory qualification.

The Merlin execution checkout remains fixed at `296fd04c0`. Documentation/audit handoff is on `codex/cosmology-linear`, commit `9506f1e52`; it must not be confused with the frozen executable source. Its final integrity audit verifies the canonical 18-run matrix, all 1,355 check identities and qualification flags, 254 comparison records, eight time-refinement records, 162 numerical archives and 268 file hashes. The report SHA-256 is

`82cb91aae41986c7655a742e71c59150a175f926db1a5e1d2d2709a9ae8156b2`.

The audit passes while scientific acceptance remains false. Its role is completeness, provenance and internal consistency, not an independent recalculation of all particle physics. The controller's exit 1 records the 116 scientific failures; there is no execution or disk error in G. Full tracked-source and executable manifests are also unchanged after plotting. Numerical NPZ archives preserve exact stored arrays and their digests, but do not reconstruct the original CSV byte formatting; the earlier gzip-based E archive does retain original CSV bytes. Large data remain on Merlin rather than being duplicated with this manuscript.

ZB is a separate follow-up campaign, not an extension of G's crossed 32/64-resolution matrix. Its complete analysis report SHA-256 is listed with the campaign record SHA-256 below.

Analysis report: `b345da2fbbf22e9a57091a06352be15d`
`a72d6296809502a571fdc130453f6122`

Campaign record: 340b497e334d50767bf83020185650e3
28ec644db498e3b0b2988d7d63ce0129

The plot manifest verifies the PNG, SVG, movie, analysis report and campaign hashes. A separate copy of the plot manifest is preserved at

`zeldovich-fastpm-plot-manifest.json`.

Particle snapshots and the machine-readable spectral arrays remain on Merlin and are not bundled with this manuscript.

GT is the subsequent single-realization GADGET-2 comparison described in Section A.10. The plot-manifest sidecar listed in Table A.6 records and rechecks the exact report, campaign, plot-source, PNG, SVG and machine-readable plot-value hashes. The paper’s copied PNG has the same digest as the rendered figure; the snapshot and full analysis remain on Merlin.

Analysis report SHA-256:

8ad14e25a766665ce3f8c347c250637a
2692fd54e06f1a0ea8ac8d3f9b5605a9

Campaign record SHA-256:

9d7c0fa83907641272c2a7fa677857e4
75e897cf858349ea8c7a50fbc35cf80d

Snapshot SHA-256:

4f10e99fc39146fb5dd17aefc426d07f
229df515a4a2dd458eef8345a7df74b5

Paper PNG SHA-256:

5890ad8f315e9cf2fe0979aa7ac02755
a49562a4be338b1a1d74e8747b4948f6

Together, the CPU campaigns test background/IC conventions, discrete forces, bounded trajectories and nonlinear PM comparisons, and MPI reproducibility. Limits include unresolved local pancake gradients, retained strict mass-sum failures, and insufficient Gaussian resolution robustness. Halo statistics, GPU and multi-node performance remain unqualified. Larger crossed resolutions, independent seeds, 2LPT/transient studies, an expanding-universe energy diagnostic, and back-end/scaling tests are subsequent work. A systematic force- and step-matched study is still required to isolate the TreePM/plain-PM and integrator contributions seen in GT. ZB’s very close matched PM spectrum does not remove any S/G resolution failure or supply a converged science band; nor does the single-resolution GT comparison qualify a continuum solution. No failed threshold is retroactively relaxed, and no precision-cosmology or exascale claim follows from the present CPU campaigns.

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